

Tuesday

candidate itemset	1 itemset	2 itemset	3 itemset
{M}	3		
{O}	3		
{N}	2 (discard ≤ 3)		
{K}	5		
{E}	4		
{Y}	3		
{D}	1 (discard ≤ 3)		
{A}	1 (discard ≤ 3)		
{U}	1 (discard ≤ 3)		
{C}	2 (discard ≤ 3)		
{I}	1 (discard ≤ 3)		

itemset	1 itemset	2 itemset	3 itemset
{M}	3		
{O}	3		
{K}	5		
{E}	4		
{Y}	3		

candidate itemset	2 itemset	3 itemset
{M, O}	1 (discard)	
{M, K}	3	
{M, E}	2 (discard)	
{M, Y}	2 (discard)	
{O, K}	3	
{O, E}	3	
{O, Y}	2 (discard)	
{K, E}	4	
{K, Y}	3	
{E, Y}	2 (discard)	

itemset	1 itemset	2 itemset	3 itemset
{M, K}	3		
{O, K}	3		
{O, E}	3		
{K, E}	4		
{K, Y}	3		

candidate itemset	3 itemset	4 itemset
{O, K, Y}	2 (discard)	
{K, E, Y}	2 (discard)	
{M, K, O}	1 (discard)	
{M, K, E}	2 (discard)	
{M, K, Y}	2 (discard)	
{O, K, E}	3	

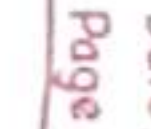
conclusion:-
The frequent 3 itemset are {O, K, E}.

→ FP Growth.

itemset	support count	item support count
{M}	3	
{O}	3	
{N}	2 (discard)	
{K}	5	
{E}	4	
{Y}	3	
{D}	1 (discard)	
{A}	1 (discard)	
{U}	1 (discard)	
{C}	2 (discard)	
{I}	1 (discard)	

item	support count	ordered items
{K}	5	K E M O Y
{E}	4	K E O Y
{M}	3	K E M
{O}	3	K M Y
{Y}	3	K E O

(leave complete steps for exam).
→ now draw final tree.

Page

Page

Page

2/4/26

2/4/26

2/4/26

2/4/26

2/4/26

2/4/26

2/4/26

2/4/26

2/4/26

2/4/26

2/4/26

2/4/26

2/4/26

2/4/26

2/4/26

2/4/26

Monday

1) Beverage preferences among a group of 1000 people.

	Coffee	Tea
only	150	50
S & C	600	150
as net	800	200

enough
for any
evaluation
support = $\frac{\sigma^2(XUY)}{N}$
Interest Factor = $\frac{150}{200 \times 800}$

Confidence = $\frac{\sigma^2(XUY)}{\sigma^2(X)}$

Support = $\frac{150}{1000} = 15\%$

Confidence = $\frac{150}{900} = 0.15 = 15\%$

→ It is negatively related (< 1)

2) Honey & Tea.

Support = 10%
Confidence = 50%

3) Interest Factors

$$\frac{I(A,B)}{S(A) \times S(B)} = \frac{N f_{11}}{f_{1+} f_{+1}}$$

→ Interest factor = $\frac{0.1}{(0.2)(0.2)} = \frac{0.1}{0.04} = 2.5$

→ Positively related (> 1).

- If equal to 1 ($= 1$) independent.
- If < 1 negatively related.
- If > 1 positively related.

3) Correlation analysis:

$$\phi = \frac{f_{11} f_{00} - f_{01} f_{10}}{\sqrt{f_{1+} f_{+1} f_{0+} f_{+0}}}$$

→ Correlation analysis is one of the most popular techniques for analyzing relationships between a pair of variables.

$$\phi = \frac{(150 \times 150) - (650 \times 50)}{\sqrt{200 \times 800 \times 800 \times 200}}$$

$$= \frac{22500 - 32500}{\sqrt{160000} \times (160000)}$$

$$= \frac{-10000}{\sqrt{25600000000}}$$

$$= \frac{-10000}{160000} = -\frac{1}{16} = -0.0625$$

< than 0 → Negatively related.

0.5847 Therefore, truly related.

Tuesday

④ IS Measure:-

$$IS(A,B) = \sqrt{IS(A,B) \times S(A,B)} = \frac{S(A,B)}{\sqrt{f_{11} + f_{1+} + f_{+1} + f_{++}}}$$

$$Tea \& Coffee = 0.375$$

$$Tea \& Honey = 0.6455$$

apriori / 2 statements use should write
problems $\rightarrow$ write reason (why discarded)

TD growth

all steps should be written.

if only construction $\rightarrow$ no table is not required

• Mention TD's on every steps.

Tuesday

UNIT-4

CLASSIFICATION

*) Classification:-

$\rightarrow$ classification is a task of learning a target function f that maps each attribute set x to one of the predefined class labels y .
eg: spam filter and its classification

$\rightarrow$ 2 types

- 1) Descriptive modeling. (explanatory tool)
- 2) Predictive modeling.

illustration of classification model:-

Confusion Matrix:-

	Predicted class	
	class = 1	class = 0
Actual class = 1	f_{11}	f_{10}
Actual class = 0	f_{01}	f_{00}

$$\text{accuracy} = \frac{\# \text{ correct predictions}}{\text{total \# of predictions}} = \frac{f_{11} + f_{00}}{f_{11} + f_{10} + f_{01} + f_{00}}$$

$$\text{Error rate} = \frac{\# \text{ wrong predictions}}{\text{total \# of predictions}} = \frac{f_{10} + f_{01}}{f_{11} + f_{10} + f_{01} + f_{00}}$$

①

Actual \ Predicted	Pass	Fail
Pass	8	2
Fail	3	7

$$\rightarrow \text{accuracy} = \frac{8+7}{8+2+3+7} = \frac{15}{20} = \frac{3}{4} = 75\%$$

$$\text{error rate} = \frac{2+3}{8+2+3+7} = \frac{5}{20} = \frac{1}{4} = 25\%$$

Thursday

Types of classifiers

- 1) Decision tree classifier.
- 2) Rule based classifier.
- 3) Nearest Neighbor classifier.
- 4) Bayesian classifier.

1) Decision tree classifier

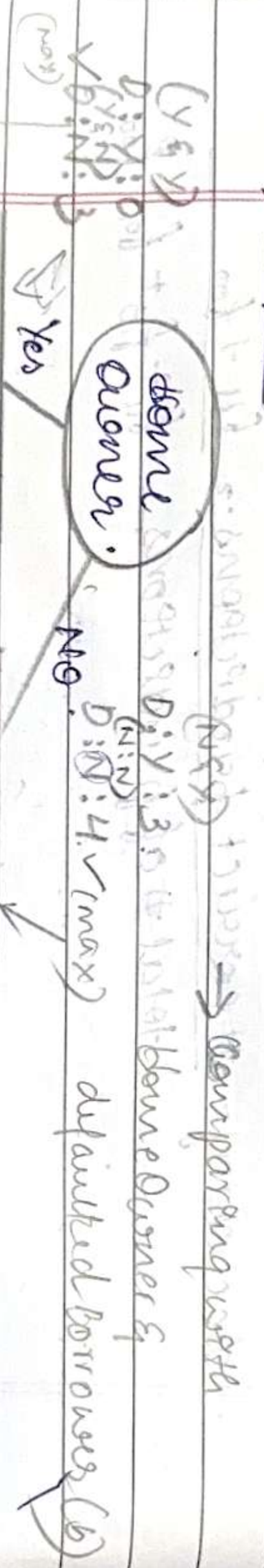
How to build a decision tree.
 $\rightarrow$ Short algorithm used to construct decision tree.

Step 1.

Defaulted = No. $\rightarrow$ 2 outcome $\rightarrow$ Yes & No

$\rightarrow$ we have given no because numerical value is "No"

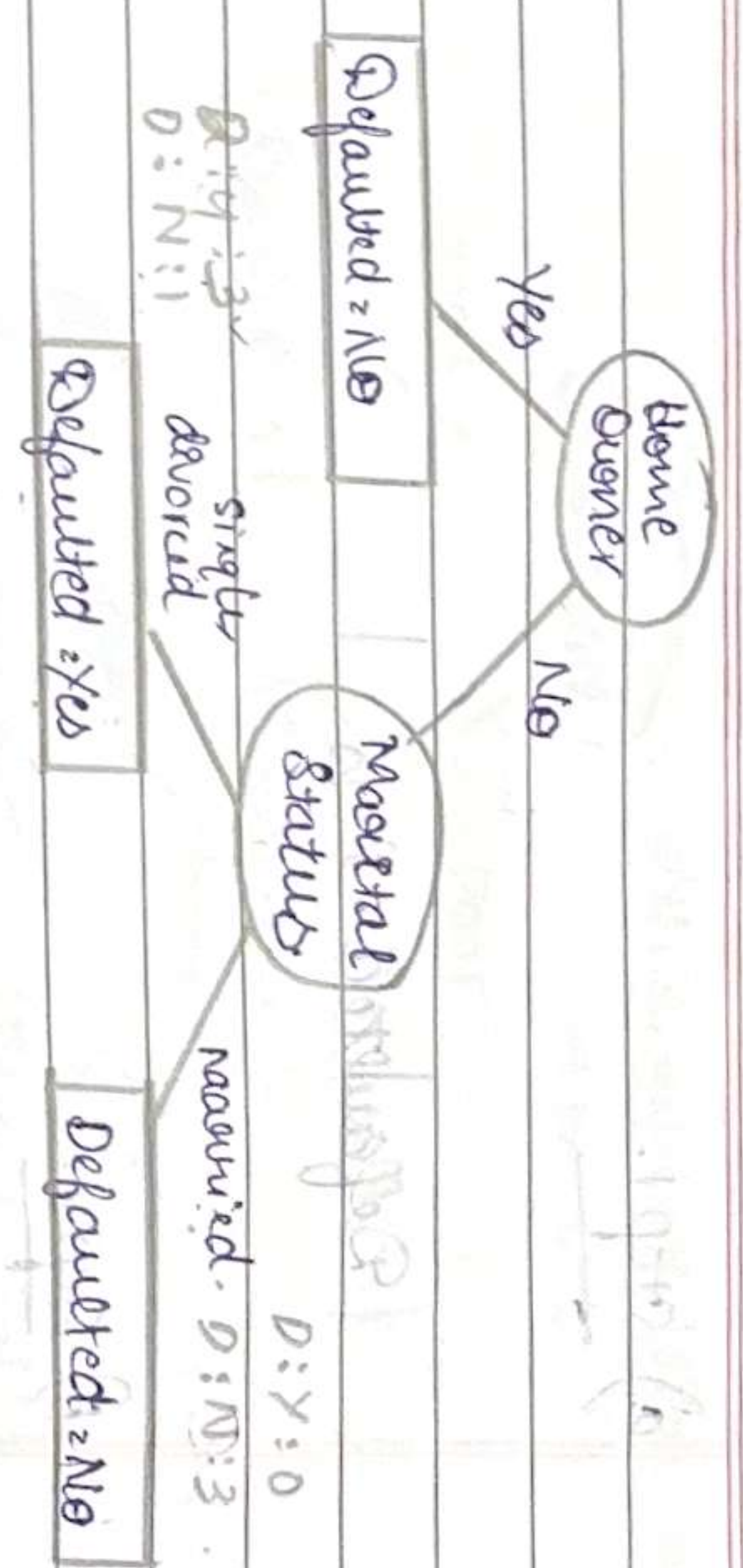
Step 2.



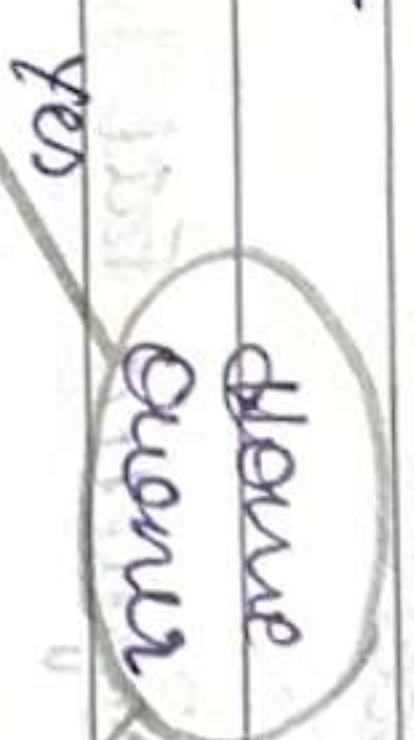
Step 3.

$\rightarrow$ D $\rightarrow$ Y H $\rightarrow$ N $\rightarrow$ single/divorced $\rightarrow$ D: Y: 3
 D $\rightarrow$ N H $\rightarrow$ Y $\rightarrow$ 24 D: N: 1
 D $\rightarrow$ Y H $\rightarrow$ N $\rightarrow$ married D: Y: 0
 D $\rightarrow$ N H $\rightarrow$ N $\rightarrow$ 3 D: N: 3

Step 4



Defaulted = No

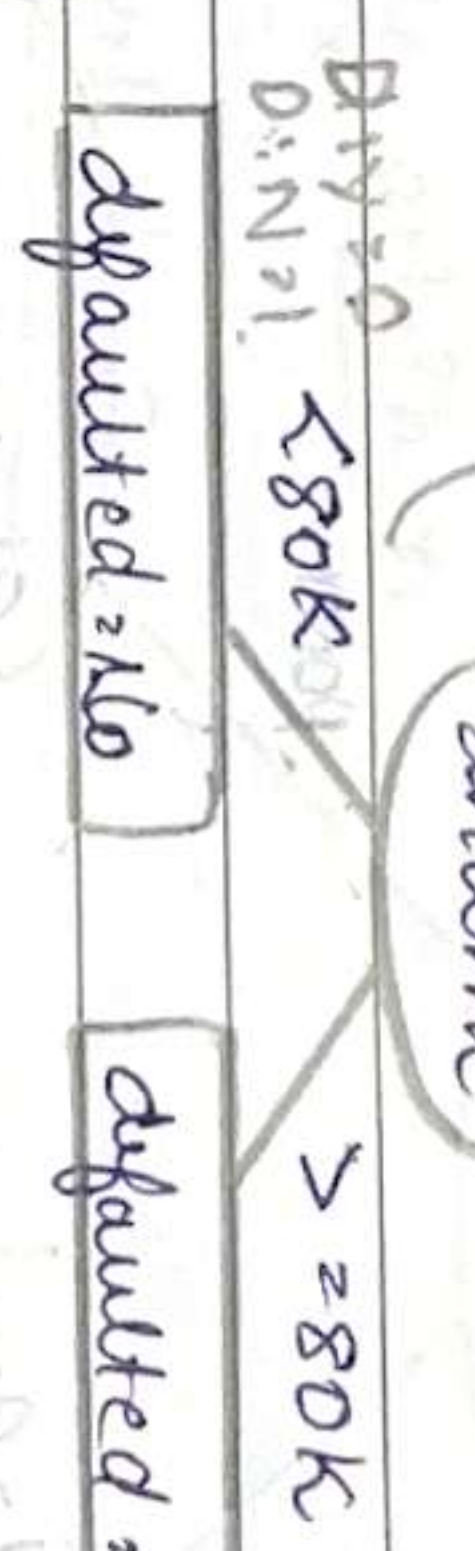


Marital Status

single, divorced

married

Defaulted = No



Student	Attendance	study hours	Internal Marks	Result
S1	good	high	high	pass
S2	good	high	high	pass
S3	poor	high	high	pass
S4	good	low	low	pass
S5	poor	low	low	fail
S6	poor	low	low	fail
S7	good	high	low	pass
S8	good	low	high	pass
S9	poor	high	low	fail
S10	poor	low	high	fail

a) Step 1.

Result & Para

b) step 2:

Attendance

D/P 25
D/P 20
good

Results:

Reset z-Tail

$D_{12} = 1$
 $D_{13} = 1$
 $D_{14} = 1$
 $D_{15} = 1$
 $D_{16} = 1$
 $D_{17} = 1$
 $D_{18} = 1$
 $D_{19} = 1$
 $D_{20} = 1$
 $D_{21} = 1$
 $D_{22} = 1$
 $D_{23} = 1$
 $D_{24} = 1$
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 $D_{93} = 1$
 $D_{94} = 1$
 $D_{95} = 1$
 $D_{96} = 1$
 $D_{97} = 1$
 $D_{98} = 1$
 $D_{99} = 1$

e) Step 3.

Attendance.

good.

Result - 2 Pass

Study hours

$P_{17}, P_{20} = 1$

Rough

$P, L, \text{Prnts} = 0$
 $\text{len } P, L, \text{Prnts} = 3$

Subject is Pans

Result = Fail -

g) Step 4.



10090 500

29 Nov

→ we split because of impurity.

→ if everything → Yes → every class

1/2

good

→ we get better side o
then shop.

Result-2 Para

Styden hovers

high.

Internal
Marks:

Result = Fail

$\rho, H, H, \rho_{\text{avg}} = 1$

DH-H; Tail 20 High

Result = Para

Result = Fail

and or Algorithm (short algorithm)

As subset $\rightarrow$

successive function (σ -D.F. $\langle E, F \rangle$)

draw a decision tree looking at

of Binary attribute.

d) Ordinal Attribute

*) continuous attrition

Apr 10, 11, F.P. will generate an explanation $\rightarrow$ task.

classmate

1) Let D1 & D2 be the documents

D1 = Java, Programming, Python, Programming, Data

D2 = Programming, Learning, Java, Data, Learning

1) Represent both documents as term-frequency vectors

→ 2)

	Java	programming	python	Data	Learning
D1	1	2	1	1	0
D2	1	1	0	1	2

2) Find the cosine similarity between D1 & D2

2) cosine similarity

$$CS = \frac{x \cdot y}{\|x\| \|y\|}$$

$$\|x\| = \sqrt{1+2+1+1+0}$$

$$\text{where } x \cdot y = \sum_{k=1}^n x_k y_k$$

$$\|x\| = \sqrt{\text{sum of } x^2} = \sqrt{1+2+1+1+0}$$

$$\|y\| = \sqrt{\text{sum of } y^2} = \sqrt{1+1+0+1+2}$$

$$x \cdot y = (1 \times 1) + (2 \times 1) + (1 \times 0) + (1 \times 1) + (0 \times 2)$$

$$= 1 + 2 + 0 + 1 + 0$$

$$= 4$$

$$\|x\| = \sqrt{1+2+1+1+0}$$

$$= \sqrt{4} = 2$$

$$\|y\| = \sqrt{1+1+0+1+2}$$

$$= \sqrt{5} = 2.236$$

$$CS = \frac{x \cdot y}{\|x\| \|y\|} = \frac{4}{2 \times 2.236}$$

$$\|y\| = \sqrt{1+1+0+1+2}$$

$$= 2$$

$$(2.64 \times 2.64)$$

$$= 4$$

$$6.9696$$

$$\cos(x, y) = 0.5739$$

∴ There is 50% similarity between 2 documents

2) 2 customers purchased items from a store

Customer C1: rice, sugar, tea, milk

Customer C2: rice, coffee, milk, biscuits

1) represent C1 & C2 using binary vectors

2) Calculate the sample matching coefficient

3) Calculate the Jaccard coefficient

	rice	sugar	tea	milk	coffee	biscuits
C1	1	1	1	1	0	0
C2	1	0	0	1	1	1

3) SMC = no. of matching attributes / no. of attributes

$$\text{no. of attributes}$$

$$= \frac{f_{11} + f_{10} + f_{01} + f_{00}}{f_{11} + f_{10} + f_{01} + f_{00}}$$

$$f_{00} = 0$$

$$f_{01} = 2$$

$$f_{10} = 2$$

$$f_{11} = 2$$

$$SMC = 0 + 2$$

$$2 + 2 + 2 + 0$$

$$= \frac{2}{6} = 0.33$$

$$J_2 = \frac{f_{11}}{f_{10} + f_{11}} = \frac{2}{2+2} = \frac{2}{4} = 0.5$$

$$J_3 = \frac{f_{11}}{f_{10} + f_{11}} = \frac{2}{2+2} = \frac{2}{4} = 0.5$$

3) Classifier produced the following results.

	Predicted class 1	Predicted class 2
Actual class 1	50	10
Actual class 2	5	35

9) Calculate accuracy and error rate

$$\Rightarrow \text{accuracy} = \frac{f_{11} + f_{22}}{f_{10} + f_{11} + f_{20} + f_{21} + f_{22}}$$

$$= \frac{50 + 35}{50 + 10 + 5 + 35} = \frac{85}{100} = 0.85 = 85\%$$

$$\text{Error rate} = \frac{f_{10} + f_{21}}{f_{10} + f_{11} + f_{20} + f_{21} + f_{22}} = \frac{10 + 5}{100} = 0.15 = 15\%$$

$$= \frac{15}{100} = 0.15 = 15\%$$

Friday

3)

Nearest Neighbor Classifier (KNN) :- (No question, structure, we should give eq)

Majority voting formula -

$$y' = \text{argmax}_{i \in D_z} I(y_i, y)$$

Name	durability	strength	class	distance
Type-1	F	F	Bad	4
Type-2	F	4	Bad	5
Type-3	3	4	Good	3
Type-4	1	4	Good	3.605

Test Data $\Rightarrow$ acid durability = 3, and strength = F

class = ?

$$\text{For } k=3$$

$$\text{③ } \sqrt{(3-3)^2 + (4-F)^2} = \sqrt{0+1} = 1$$

$$\text{④ } \sqrt{(3-3)^2 + (4-F)^2} = \sqrt{0+1} = 1$$

$$\text{⑤ } \sqrt{(3-3)^2 + (4-F)^2} = \sqrt{0+1} = 1$$

$$\text{⑥ } \sqrt{(1-3)^2 + (4-F)^2} = \sqrt{4+1} = \sqrt{5} = 2.236$$

$$\text{⑦ } \sqrt{(1-3)^2 + (4-F)^2} = \sqrt{4+1} = \sqrt{5} = 2.236$$

There nearest neighbors are Type 1, Type 2, Type 3, Type 4.

The test data is classified as "good".

Ques #1) K-NN algorithm -> (k-nearest neighbor)

Q Given the following training dataset. Let a new data point is Age: 30, Income: 400.

Age	Income	Class
25	30	No
35	60	Yes
45	80	Yes
20	20	No

Age: 30 Income: 400

Use 1-Nearest Neighbor method to classify the new data point.

1) Euclidean distance

$$D = \sqrt{(30-25)^2 + (400-30)^2}$$

$$= \sqrt{(5)^2 + (370)^2}$$

$$= \sqrt{25 + 136900} = 11.180$$

$$D = \sqrt{(30-35)^2 + (400-60)^2}$$

$$= \sqrt{(5)^2 + (340)^2}$$

$$= \sqrt{25 + 115600} = 20.615$$

$$D = \sqrt{(30-45)^2 + (400-80)^2}$$

$$= \sqrt{(15)^2 + (320)^2}$$

$$= \sqrt{225 + 102400} = 31.931$$

$$D = \sqrt{(30-20)^2 + (400-20)^2}$$

$$= \sqrt{(10)^2 + (380)^2}$$

$$= \sqrt{100 + 144400} = 37.860$$

As

Our nearest neighbor is Type 1

Majority voting formula.

$$y' = \arg \min_x \sum (x_i y_i) \text{ E.D.I (v-y)}$$

Identify rules burden
-> accuracy, coverage & priority

-> The test data is classified as "No"

2) Rule Based classifier (if-then rules)

eg: If (Age > 30) & (Income > 400) -> Yes.

Each classification rule can be expressed in the following way.

$$if(\text{condition}) \rightarrow y_i$$

-> Syntax

$$\text{condition} = (A1 \text{ op } v1) \wedge (A2 \text{ op } v2) \wedge \dots \wedge (Ak \text{ op } vk)$$

where (A_i, v_i) is an attribute-value pair and op is a logical operator chosen from the set {&, |, <, >, <=, >=, <=, >=}

eg -> Left hand side of the rule is called the rule antecedent (part of precondition).

eg a) Rule coverage and accuracy -

-> Coverage of a rule:

Fraction of records satisfy the antecedent of a rule.

Accuracy of a rule: Fraction of records that satisfy the antecedent that also satisfy the consequent of a rule.

$$\text{Coverage}(x) = \frac{|A|}{|D|} \quad \text{Accuracy}(x) = \frac{|A \cap y|}{|A|}$$

(condition -> class)

Tuesday

eg: (status = single) $\rightarrow$ NO.

coverage(x) = $\frac{|A|}{|D|}$ 1/ total no. of records that satisfy the rule antecedent.

$$= \frac{4}{28} = 0.14 = 14\%$$

accuracy = $\frac{|A \cap y|}{|A|}$ // total no. of records that satisfy rule antecedent

$$= \frac{21}{42} = 0.5 = 50\%$$

	Age	Gender	Married	Occupation	Education	Income	Household Income	Number of Children	Number of Siblings	Number of Parents	Number of Grandchildren
R3	30	Male	Yes	Engineer	High School	50000	100000	2	3	4	5
	35	Female	No	Teacher	College	40000	80000	1	2	3	4
	40	Male	Yes	Doctor	University	60000	120000	3	4	5	6
R4	45	Female	No	Lawyer	High School	70000	140000	2	3	4	5
	50	Male	Yes	Engineer	College	80000	160000	3	4	5	6
	55	Female	No	Teacher	University	90000	180000	2	3	4	5
R5	60	Male	Yes	Doctor	High School	100000	200000	3	4	5	6
	65	Female	No	Lawyer	College	110000	220000	2	3	4	5
	70	Male	Yes	Engineer	University	120000	240000	3	4	5	6
R6	75	Female	No	Teacher	High School	130000	260000	2	3	4	5
	80	Male	Yes	Doctor	College	140000	280000	3	4	5	6
	85	Female	No	Lawyer	University	150000	300000	2	3	4	5
R7	90	Male	Yes	Engineer	High School	160000	320000	3	4	5	6
	95	Female	No	Teacher	College	170000	340000	2	3	4	5
	100	Male	Yes	Doctor	University	180000	360000	3	4	5	6
R8	105	Female	No	Lawyer	High School	190000	380000	2	3	4	5
	110	Male	Yes	Engineer	College	200000	400000	3	4	5	6
	115	Female	No	Teacher	University	210000	420000	2	3	4	5
R9	120	Male	Yes	Doctor	High School	220000	440000	3	4	5	6
	125	Female	No	Lawyer	College	230000	460000	2	3	4	5
	130	Male	Yes	Engineer	University	240000	480000	3	4	5	6
R10	135	Female	No	Teacher	High School	250000	500000	2	3	4	5
	140	Male	Yes	Doctor	College	260000	520000	3	4	5	6
	145	Female	No	Lawyer	University	270000	540000	2	3	4	5
R11	150	Male	Yes	Engineer	High School	280000	560000	3	4	5	6
	155	Female	No	Teacher	College	290000	580000	2	3	4	5
	160	Male	Yes	Doctor	University	300000	600000	3	4	5	6
R12	165	Female	No	Lawyer	High School	310000	620000	2	3	4	5
	170	Male	Yes	Engineer	College	320000	640000	3	4	5	6
	175	Female	No	Teacher	University	330000	660000	2	3	4	5
R13	180	Male	Yes	Doctor	High School	340000	680000	3	4	5	6
	185	Female	No	Lawyer	College	350000	700000	2	3	4	5
	190	Male	Yes	Engineer	University	360000	720000	3	4	5	6
R14	195	Female	No	Teacher	High School	370000	740000	2	3	4	5
	200	Male	Yes	Doctor	College	380000	760000	3	4	5	6
	205	Female	No	Lawyer	University	390000	780000	2	3	4	5
R15	210	Male	Yes	Engineer	High School	400000	800000	3	4	5	6
	215	Female	No	Teacher	College	410000	820000	2	3	4	5
	220	Male	Yes	Doctor	University	420000	840000	3	4	5	6
R16	225	Female	No	Lawyer	High School	430000	860000	2	3	4	5
	230	Male	Yes	Engineer	College	440000	880000	3	4	5	6
	235	Female	No	Teacher	University	450000	900000	2	3	4	5
R17	240	Male	Yes	Doctor	High School	460000	920000	3	4	5	6
	245	Female	No	Lawyer	College	470000	940000	2	3	4	5
	250	Male	Yes	Engineer	University	480000	960000	3	4	5	6
R18	255	Female	No	Teacher	High School	490000	980000	2	3	4	5
	260	Male	Yes	Doctor	College	500000	1000000	3	4	5	6
	265	Female	No	Lawyer	University	510000	1020000	2	3	4	5
R19	270	Male	Yes	Engineer	High School	520000	1040000	3	4	5	6
	275	Female	No	Teacher	College	530000	1060000	2	3	4	5
	280	Male	Yes	Doctor	University	540000	1080000	3	4	5	6
R20	285	Female	No	Lawyer	High School	550000	1100000	2	3	4	5
	290	Male	Yes	Engineer	College	560000	1120000	3	4	5	6
	295	Female	No	Teacher	University	570000	1140000	2	3	4	5
R21	300	Male	Yes	Doctor	High School	580000	1160000	3	4	5	6
	305	Female	No	Lawyer	College	590000	1180000	2	3	4	5
	310	Male	Yes	Engineer	University	600000	1200000	3	4	5	6
R22	315	Female	No	Teacher	High School	610000	1220000	2	3	4	5
	320	Male	Yes	Doctor	College	620000	1240000	3	4	5	6
	325	Female	No	Lawyer	University	630000	1260000	2	3	4	5
R23	330	Male	Yes	Engineer	High School	640000	1280000	3	4	5	6
	335	Female	No	Teacher	College	650000	1300000	2	3	4	5
	340	Male	Yes	Doctor	University	660000	1320000	3	4	5	6
R24	345	Female	No	Lawyer	High School	670000	1340000	2	3	4	5
	350	Male	Yes	Engineer	College	680000	1360000	3	4	5	6
	355	Female	No	Teacher	University	690000	1380000	2	3	4	5
R25	360	Male	Yes	Doctor	High School	700000	1400000	3	4	5	6
	365	Female	No	Lawyer	College	710000	1420000	2	3	4	5
	370	Male	Yes	Engineer	University	720000	1440000	3	4	5	6
R26	375	Female	No	Teacher	High School	730000	1460000	2	3	4	5
	380	Male	Yes	Doctor	College	740000	1480000	3	4	5	6
	385	Female	No	Lawyer	University	750000	1500000	2	3	4	5
R27	390	Male	Yes	Engineer	High School	760000	1520000	3	4	5	6
	395	Female	No	Teacher	College	770000	1540000	2	3	4	5
	400	Male	Yes	Doctor	University	780000	1560000	3	4	5	6
R28	405	Female	No	Lawyer	High School	790000	1580000	2	3	4	5
	410	Male	Yes	Engineer	College	800000	1600000	3	4	5	6
	415	Female	No	Teacher	University	810000	1620000	2	3	4	5
R29	420	Male	Yes	Doctor	High School	820000	1640000	3	4	5	6
	425	Female	No	Lawyer	College	830000	1660000	2	3	4	5
	430	Male	Yes	Engineer	University	840000	1680000	3	4	5	6
R30	435	Female	No	Teacher	High School	850000	1700000	2	3	4	5
	440	Male	Yes	Doctor	College	860000	1720000	3	4	5	6
	445	Female	No	Lawyer	University	870000	1740000	2	3	4	5
R31	450	Male	Yes	Engineer	High School	880000	1760000	3	4	5	6
	455	Female	No	Teacher	College	890000	1780000	2	3	4	5
	460	Male	Yes	Doctor	University	900000	1800000	3	4	5	6
R32	465	Female	No	Lawyer	High School	910000	1820000	2	3	4	5
	470	Male	Yes	Engineer	College	920000	1840000	3	4	5	6
	475	Female	No	Teacher	University	930000	1860000	2	3	4	5
R33	480	Male	Yes	Doctor	High School	940000	1880000	3	4	5	6
	485	Female	No	Lawyer	College	950000	1900000	2	3	4	5
	490	Male	Yes	Engineer	University	960000	1920000	3	4	5	6
R34	495	Female	No	Teacher	High School	970000	1940000	2	3	4	5
	500	Male	Yes	Doctor	College	980000	1960000	3	4	5	6
	505	Female	No	Lawyer	University	990000	1980000	2	3	4	5
R35	510	Male	Yes	Engineer	High School	1000000	2000000	3	4	5	6
	515	Female	No	Teacher	College	1010000	2020000	2	3	4	5
	520	Male	Yes	Doctor	University	1020000	2040000	3	4	5	6
R36	525	Female	No	Lawyer	High School	1030000	2060000	2	3	4	5
	530	Male	Yes	Engineer	College	1040000	2080000	3	4	5	6
	535	Female	No	Teacher	University	1050000	2100000	2	3	4	5
R37	540	Male	Yes	Doctor	High School	1060000	2120000	3	4	5	6
	545	Female	No	Lawyer	College	1070000	2140000	2	3	4	5
	550	Male	Yes	Engineer	University	1080000	2160000	3	4	5	6
R38	555	Female	No	Teacher	High School	1090000	2180000	2	3	4	5
	560	Male	Yes	Doctor	College	1100000	2200000	3	4	5	6
	565	Female	No	Lawyer	University	1110000	2220000	2	3	4	5
R39	570	Male	Yes	Engineer	High School	1120000	2240000	3	4	5	6
	575	Female	No	Teacher	College	1130000	2260000	2	3	4	5
	580	Male	Yes	Doctor	University	1140000	2280000	3	4	5	6
R40	585	Female	No	Lawyer	High School	1150000	2300000	2	3	4	5
	590	Male	Yes	Engineer	College	1160000	2320000	3	4	5	6
	595	Female	No	Teacher	University	1170000	2340000	2	3	4	5
R41	600	Male	Yes	Doctor	High School	1180000	2360000	3	4	5	6
	605	Female	No	Lawyer	College	1190000	2380000	2	3	4	5
	610	Male	Yes	Engineer	University	1200000	2400000	3	4	5	6
R42	615	Female	No	Teacher	High School	1210000	2420000	2	3	4	5
	620	Male	Yes	Doctor	College	1220000	2440000	3	4	5	6
	625	Female	No	Lawyer	University	1230000	2460000	2	3	4	5
R43	630	Male	Yes	Engineer	High School	1240000	2480000	3	4	5	6
	635	Female	No	Teacher	College	1250000	2500000	2	3	4	5
	640	Male	Yes	Doctor	University	1260000	2520000	3	4	5	6
R44	645	Female	No	Lawyer	High School	1270000	2540000	2	3	4	5
	650	Male	Yes	Engineer	College	1280000	2560000	3	4	5	6
	655	Female	No	Teacher	University	1290000	2580000	2	3	4	5
R45	660	Male	Yes	Doctor	High School	1300000	2600000	3	4	5	6
	665	Female	No	Lawyer	College	1310000	2620000	2	3	4	5
	670	Male	Yes	Engineer	University	1320000	2640000	3	4	5	6
R46	675	Female	No	Teacher	High School	1330000	2660000	2	3	4	5
	680	Male	Yes	Doctor	College	1340000	2680000	3	4	5	6
	685	Female	No	Lawyer	University	1350000	2700000	2	3	4	5
R47	690	Male	Yes	Engineer	High School	1360000	2720000	3	4	5	6
	695	Female	No	Teacher	College	1370000	2740000	2	3	4	5
	700	Male	Yes	Doctor	University	1380000	2760000	3	4	5	6
R48	705	Female	No	Lawyer	High School	1390000	2780000	2	3	4	5
	710	Male	Yes	Engineer	College	1400000	2800000	3	4	5	6
	715	Female	No	Teacher	University	1410000	2820000	2	3	4	5
R49	720	Male	Yes	Doctor	High School	1420000	2840000	3	4	5	6
	725	Female	No	Lawyer	College	1430000					

$$= \frac{4/7 \times 7/10}{4/10} = 1.00$$

② defaulted borrower: Yes | marital status = married
if marital status = married | defaulted borrower: Yes X

defaulted borrower: Yes
if marital status = married

$$= \frac{0/3 \times 3/10}{4/10}$$

$$= 0 \times 3/10$$

$$= \frac{0}{4/10} = 0$$

$$= \frac{0}{4/10} = 0$$

Conclusion -

since P(Defaulted borrower = No | marital status = married) > P(Defaulted borrower = Yes | marital status = married)

the tuple with attribute marital status = married is classified as default borrower = No.

③ ^{Act 50} ~~group~~ * Estimating predictive accuracy of classification method -

1) Holdout method / Test Sample method:-

2) Random sub-sampling method.

3) k-fold cross validation method.

4) Boot strap method.

Important

group *

Characteristics of decision tree induction.

1) Non-parametric approach. (not fixed)
NP-complete. Problem (non-Deterministic polynomial time)

2) Computationally expensive. (exp.)
easy to interpret. (easy to write)

3) Top-down, recursive (construction from top)

4) Redundant attributes does not affect the accuracy (applicability of feature selection)

5) Many algorithms like brute algorithm, CART, ID3, C4.5, SLIQ, SPIN.

group * Characteristics of Nearest Neighbor classifier.

1) Classifies based on similarity. (same type)

2) Uses distance metrics. (euclidean distance - use used)

3) Instance based. (as question and data comes, it will prepare)

4) Lazy learner. (he will not prepare and del)

5) Works for both classification and prediction. (used for both)

* Characteristics of rule-based classifiers.

1) Uses if-then rules.

2) Easy to understand.

3) Can have multiple rules for same class.

4) Order of rules may matter.

* Characteristics of a Bayesian classifier -

1) Uses Bayes Theorem.

2) Based on probability.

3) Fast and efficient.

4) Works well with high-dimensional data. (under attribution)

Tuesday

CLUSTER ANALYSIS.

UNIT-5

CLASSMATE
Date 5/5/2025
Page

51
59
54, 57
53
58

55
52
54, 510

classmate
Date
Page

* Cluster analysis groups objects based on information found in the data the dissimilarity effects and their similarities.
eg: News paper article.

k-means method: (centroid method).
used for clustering with eg.
euclidean distance measure.

→ we use Manhattan distance → it is called k-median.

Example Explain k-means method with eg and algorithm.

Student	Age	Mark1	Mark2	Mark3	C1	C2	C3	C1
S1	18	73	75	57	0	34	18	C1
S2	18	79	85	75	34	0	52	C2
S3	23	70	70	52	18	52	0	C3
S4	20	55	55	55	42	76	36	C3
S5	22	85	86	87	57	23	67	C2
S6	19	91	90	89	66	32	82	C2
S7	20	70	65	60	18	46	16	C3
S8	21	53	56	59	49	79	40	C3
S9	19	82	82	60	20	22	36	C1
S10	47	75	76	77	52	44	60	C2

Step 1: let $k=3$ (clusters).

Step 2: let seed be the first 3 students.

Student	Age	Mark1	Mark2	Mark3
S1	18	73	75	57
S2	18	79	85	75
S3	23	70	70	52

Mean value.

① $\frac{S1 + S9}{2}$

① $\frac{18 + 19}{2} = 18.5$ ② $\frac{73 + 82}{2} = 77.5$ ③ $\frac{75 + 82}{2} = 78.5$

④ $\frac{57 + 60}{2} = 58.5$

⑤ $\frac{S2, S5, S6, S10}{4}$

① $\frac{18 + 22 + 19 + 47}{4} = 26.5$ ② $\frac{79 + 85 + 91 + 75}{4} = 82.5$

③ $\frac{85 + 86 + 90 + 76}{4} = 84.3$ ④ $\frac{75 + 87 + 89 + 77}{4} = 82$

⑥ $\frac{S3, S4, S7, S8}{4}$

① $\frac{23 + 20 + 20 + 21}{4} = 21$ ② $\frac{70 + 55 + 70 + 53}{4} = 61.5$

③ $\frac{70 + 55 + 65 + 56}{4} = 61.5$ ④ $\frac{52 + 55 + 60 + 59}{4} = 56.5$

C1	18.5	77.5	78.5	58.5
C2	26.5	82.5	84.3	82
C3	21	61.5	61.5	56.5

Iteration-2. (Step 3 and 4)

		Distance from			Cluster			Allocation to the nearest cluster.		
		C1	C2	C3						
C1	18.5	77.5	78.5	58.5						
C2	26.5	82.5	84.3	82						
C3	21	62	61.5	56.6						
Student	Age	Mark1	Mark2	Mark3	C1	C2	C3			
S1	18	43	75	57	10	52.3	28			C1
S2	18	79	85	75	25	19.8	62			C2
S3	23	70	70	52	27	60.3	23			C3
S4	20	55	55	55	51	90.3	16			C3
S5	22	85	86	87	47	13.8	79			C2
S6	19	91	90	89	56	28.8	92			C2
S7	20	70	65	60	24	60.3	16			C3
S8	21	53	56	59	50	86.3	17			C3
S9	19	82	82	60	10	30.3	46			C1
S10	47	75	76	77	52	41.3	74			C2

Recluster:- Clusters have not changed.

Cluster 1 $\rightarrow$ C1 $\rightarrow$ S1, S9.

Cluster 2 $\rightarrow$ C2 $\rightarrow$ S2, S5, S6, S10.

Cluster 3 $\rightarrow$ C3 $\rightarrow$ S3, S4, S7, S8.

Example 2. (take 2 clusters)

Following are the marks of students in a class;

10, 12, 15, 30, 32, 35.

Seed 1 = 10.

Seed 2 = 30.

different types of clustering -

- 1) Hierarchical clustering Vs Partitioning clustering.
- 2) Exclusive Vs Overlapping Vs Fuzzy.
- 3) Compact Vs Partial.

Friday

agglomerative Hierarchical clustering -

$\rightarrow$ agglomerative
 $\rightarrow$ Dissolve.

Proximity measures -

- MIN
- MAX
- Group average.

Step 1

Given the following distance matrix -

	A	B	C	D
A	0	2	6	10
B	2	0	5	9
C	6	5	0	4
D	10	9	4	0

Agglomerative Hierarchical clustering (show step by step iteration)

Step 1: Initial clusters



Step 2: A and B merged because min distance = 2.

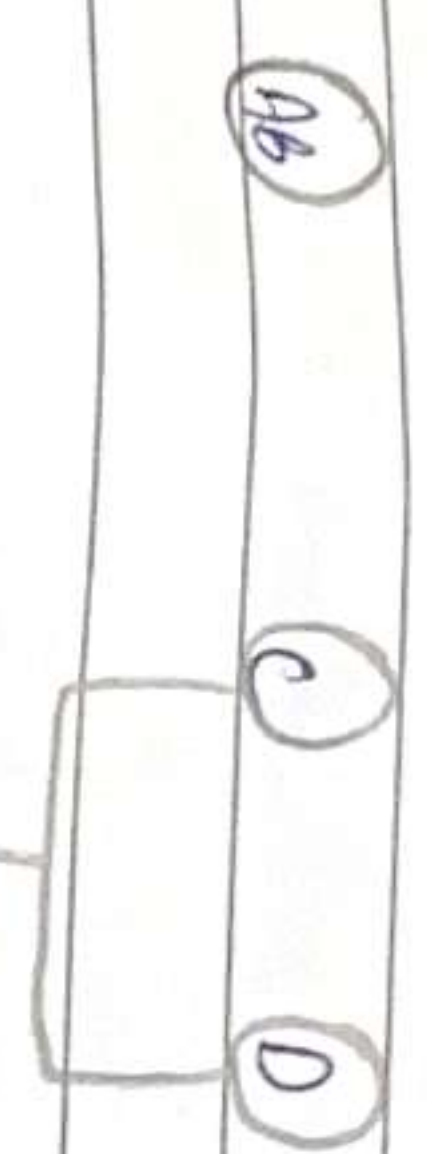


	AB	C	D
AB	0	5	9
C	5	0	4
D	9	4	0

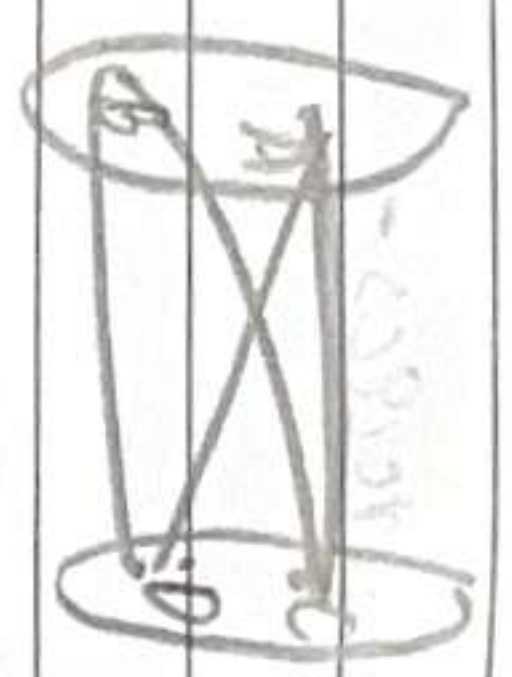


min (AC, BC) = min (AD, BD)
min (6, 5) = min (10, 9)
= 5

Step 3: Merge C and D into 4,

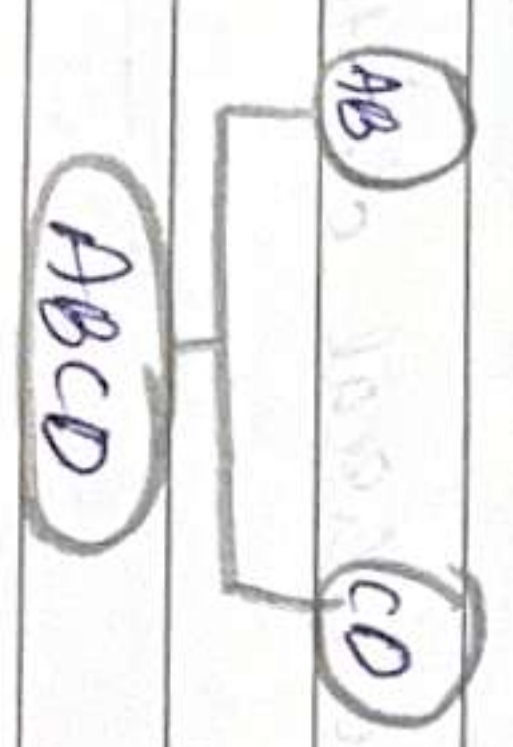


	AB	CD
AB	0	5
CD	5	0

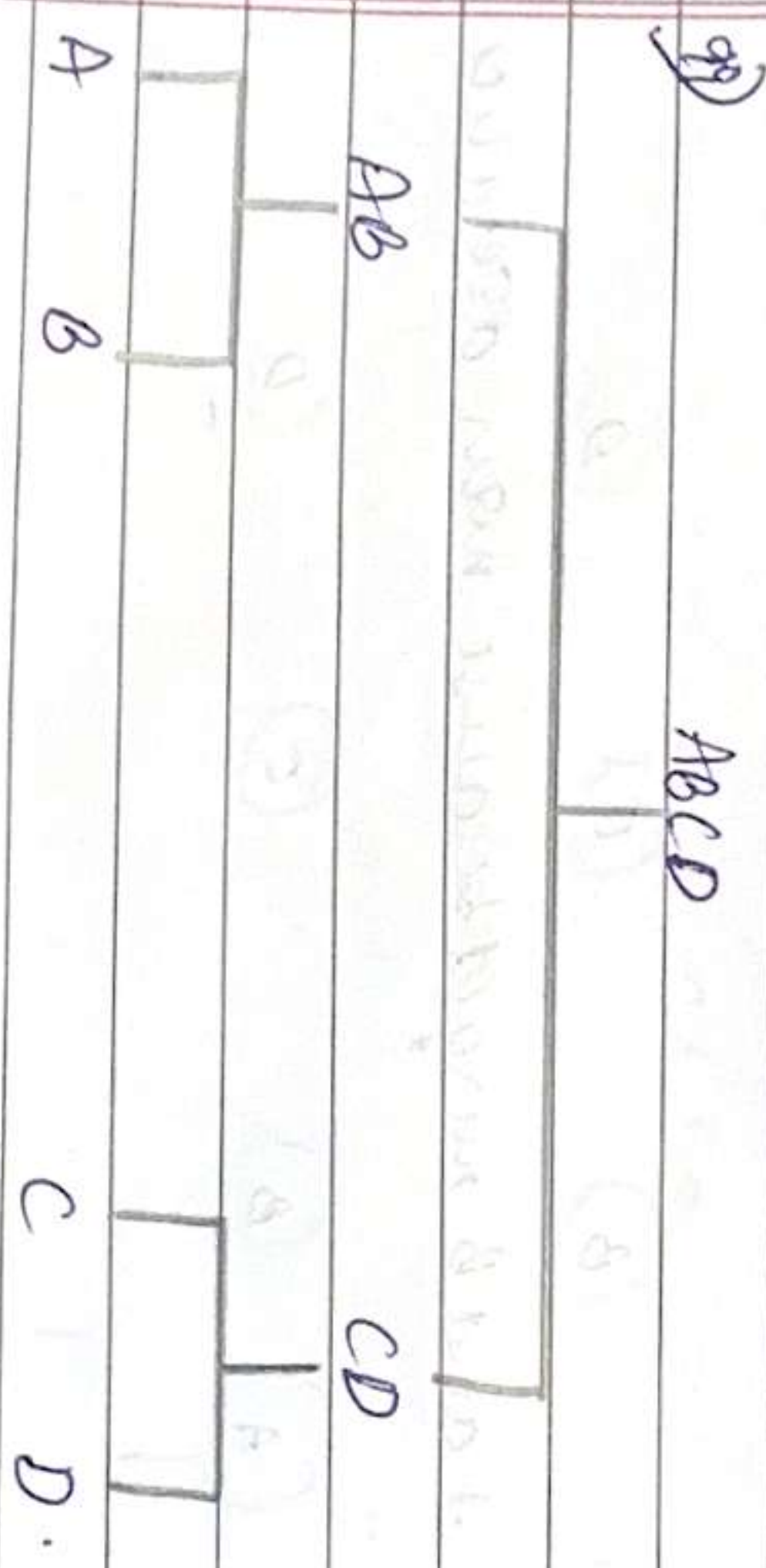


when AC, AD, BC, BD.
min 6, 10, 5, 9).
= 54

Step 4:-



9)



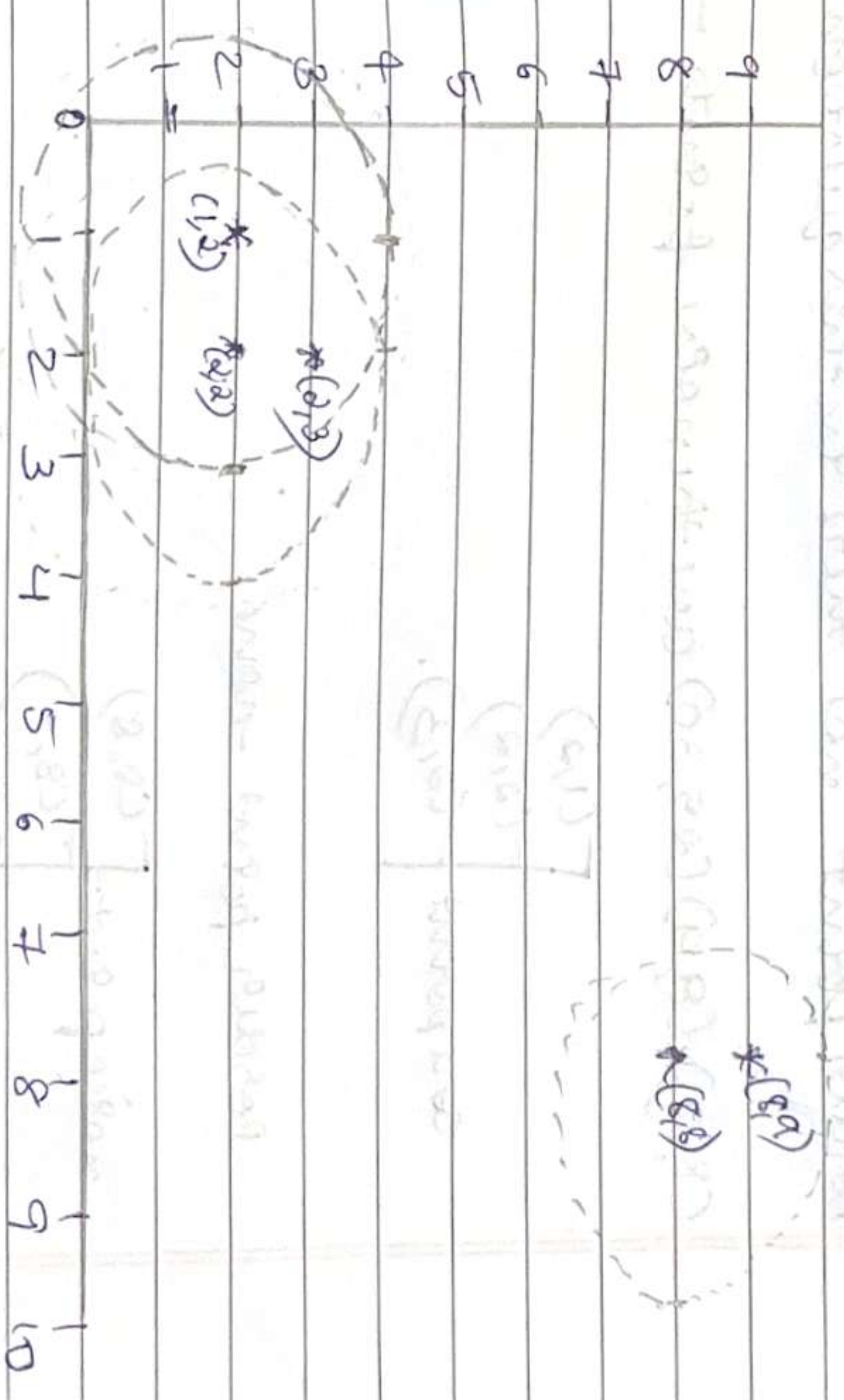
Monday

11/5/26

Explain DBSCAN Algorithm with eg.
DBSCAN (Density based spatial clustering
applications with noise).

- 1) classify the points into following categories -
- 2) core points - are in the interior points of a density based clusters.
- 3) border points - are not a core point but falls within the neighborhood of the core point.
- 4) noise point - is any point that is neither a core point nor a border point.

* Given the following dataset -
(1,2), (2,2), (2,3), (6,8), (8,9), (8,5,30).
given eps=2 and minpts=3
identify
• core points • border point • noise point



i) Point $(1,2)$ has at least 3 point within its eps distance (including itself).
 $\therefore (1,2)$ is a core point.

ii) Point $(2,2)$ has atleast 3 point within its eps distance.
 $\therefore (2,2)$ is a core point.

iii) Point $(2,3)$ has atleast 3 point within its eps distance.
 $\therefore (2,3)$ is a core point.

iv) Point $(8,8)$ is not a core point.

v) Point $(8,9)$ is not a core point.

vi) Point $(25,30)$ is not a core point.

Border point not there in this question.

$(8,9)$ $(8,9)$ $(25,30)$ are the noise points —

core point $\left[\begin{array}{l} (1,2) \\ (2,2) \\ (2,3) \end{array} \right]$.

Border point - none.

noise point $\left[\begin{array}{l} (8,8) \\ (8,9) \\ (25,30) \end{array} \right]$.